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Compressor/Diffuser/Combustor Aerodynamic Interactions in Lean Module Combustors

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The paper reports an experimental investigation into the possibility of increased interactions between combustor external aerodynamics and upstream components, e.g., prediffuser, compressor outlet guide vane (OGV), and even the compressor rotor, caused by the trend in lean module fuel injectors to larger mass flows entering the combustor cowl. To explore these component interaction effects, measurements were made on a fully annular rig comprising a single stage compressor, an advanced integrated OGV/prediffuser, followed by a dump diffuser and a generic combustor flametube with metered cowl and inner/outer annulus flows. The flow split entering the cowl was increased from 30% to 70%. The results demonstrate that, with fixed geometry, as the injector flow increases, the performance of the prediffuser and feed annuli suffer. Prediffuser losses increase and at high injector flow rates, the diffuser moves close to separation. The substantial circumferential variation in cowl flow can feed upstream and cause rotor forcing. Notable differences in performance were observed inline and between injectors at the OGV exit, suggesting that geometry changes such as an increased dump gap or nonaxisymmetric prediffuser designs may be beneficial. [DOI: 10.1115/1.2747646]

Introduction

In future aeroengine gas turbine combustor designs, lean module combustion will be essential in order to achieve required NO_x targets. Consequently, up to 70% of the flow issuing from the compressor may be required to enter the flametube through the fuel injectors, compared with (typically) 30% in current combustor designs. To accommodate this increased mass flow requirement while maintaining an acceptable pressure drop, lean module fuel injectors will be significantly larger than their conventional air-spray counterparts. This will necessitate geometry changes to the combustor external aerodynamic layout such as a deeper flametube, potentially increasing the aerodynamic loss due to increased turning within the dump region. In addition to affecting the turning losses incurred by the residual airflow entering the feed annuli surrounding the flametube, these geometry and flow split changes and the strongly 3D (circumferential) distribution of the flow into the combustor head compared to current practice are likely to increase the coupling or interaction between the combustor external aerodynamics and upstream components such as the compressor outlet guide vane (OGV)/prediffuser combination and possibly even lead to forcing of the final compressor rotor stage.

Upstream modification of turbomachinery flow due to the potential field introduced by downstream components is by no means a new phenomenon. It has led, for example, to work on optimization of the aerodynamic interaction between the fan and by-pass OGVs (Shahpar et al. [1]). Equally, the two-way interaction between the prediffuser, upstream OGV, and downstream combustor is well known. The impact of OGV wakes on the performance of a downstream located prediffuser has been fully documented (Stevens et al. [2], Barker and Carrotte [3,4]), as has the beneficial effect of the blockage caused by the downstream combustor cowl stabilizing the prediffuser flow (Klein [5]). Recent works by Barker et al. [6] and Walker et al. [7] have shown how, by adopting an integrated approach whereby the design of the OGV and prediffuser are considered simultaneously, the OGV wake flows can be manipulated to advantage in achieving improved prediffuser performance. Finally, computational fluid dynamics (CFD) simulations have also recently addressed the topic of interaction between closely coupled components such as compressor and combustor (Schlueter et al. [8]). In this work, the main effort was placed on identification of appropriate techniques for coupling the unsteady Reynolds-averaged Navier-Stokes prediction of the compressor flow with the large-eddy simulation prediction of the prediffuser/combustor combination. It seems that the coupling was essentially one way (downstream in the engine), so that although combustor cowl/fuel injector pressure and flow characteristics could influence the prediffuser, it was assumed that these did not penetrate further upstream and influence the compressor calculation. While this is certainly the design intent, and is a good assumption for present designs, the present paper is aimed at addressing the question as to how strong this upstream effect might be for lean module type geometries and injector flow splits.

The present investigation extends the study of Walker et al. [7] to consideration of combustor cowl (fuel injector) flow in addition to the compressor/prediffuser components. An experimental investigation has been undertaken to characterize the extent that the component interactions described above are modified when increasing the injector airflow from conventional levels to those expected in lean module combustors, using a combustion system layout typical of current design practice. The experimental study utilizes the same rig as used by Walker et al. [7]. This comprises a fully annular generic lean module combustor downstream of a single stage axial compressor. The OGV/prediffuser coupling is effected via an advanced integrated OGV/prediffuser geometry [4,5]. Several traverse planes are used to gather five-hole probe data that allow the flow structure to be examined through the rotor, OGV, prediffuser, and in the inner/outer annulus supply ducts, quantifying the extent of the component interactions and associated system performance. Several configurations are investigated where the injector mass flow has been increased systematically from 30% to 50% and finally 70%.

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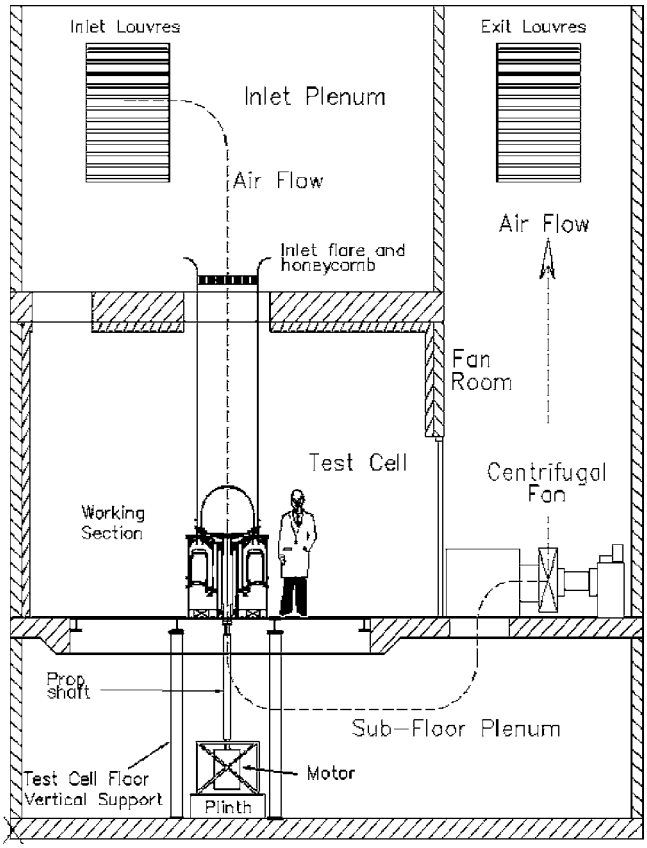

Fig. 1 Test facility

Experimental Facility

During this project, all experimental data were obtained on a low speed isothermal test facility operating at nominally atmospheric conditions. Figure 1 provides an illustrative sketch, and full details are provided in [7]. During the present study, an additional centrifugal fan was added, in series, to the system. This gave a greater flexibility in control and provided the extra drive required to overcome the increased rig pressure drop incurred by forcing a higher proportion of air through the combustor cowl.

The working section (Fig. 2) is comprised of a single stage axial compressor, an OGV prediffuser assembly, and a representation of a downstream combustion system. Geometric details on integrated guide vane (IGV), rotor, and OGV rows are given in [7]. The rotor is operated at a fixed nondimensional mass flow ($\dot{m}\sqrt{(RT)}/AP$), and speed ($N\pi D/\sqrt{(\gamma RT)}$) corresponding to a flow coefficient (V_a/U) of ϕ=0.403 and a mass flow of approximately 4 kg s^{-1}. This results in an axial velocity through the blade rows of approximately 40 ms^{-1}(Mach number 0.12) and an OGV Reynolds number of approximately 1.6×10^5. The downstream flow distribution to the feed annuli and combustor cowl is controlled by a system of throttles; the centrifugal fan is utilized as a further throttle to ensure that the axial compressor maintains the desired operating condition.

Test Geometry

Integrated OGV/Prediffuser. The OGV row has 160 vanes whose geometry includes specific lean, sweep, and camber characteristics that were determined in [6] by a process of manual CFD optimization. This achieves a spanwise blade loading variation that creates secondary flows to transport higher momentum towards the walls, thus offloading the end wall boundary layers within the prediffuser. Hence, as reported in [7], this integrated OGV (IOGV) design was able to achieve a prediffuser area ratio of 1.8 without separation. At the same nondimensional length of 2.23, this would have resulted in separation according to the standard Howard et al. [9] design chart.

Generic Flame Tube Cowl and "Fuel Injectors." The same simple flametube geometry as used previously [7] was retained in the present work. This presents the IOGV/prediffuser with a representative downstream blockage. Lean module combustors may not feature air admission ports in the annuli, hence no dilution holes were included in the flametube liner walls. The resultant generic flametube, illustrated in Fig. 3, has 20 "fuel injectors" represented by 20 simple holes, with the area of these set initially (Test 1) to match the effective area of an injector sized to take 30% of compressor discharge flow. As the mass flow through "injectors" and hence the cowl was increased, the ratio of mass flow to "injector" area was maintained by increasing the size of the holes. This meant that as the mass flow increased, the relative acceleration of the dump flow into the cowl did not change. The final configurations tested are detailed in Table 1.

Instrumentation

Probes. Aerodynamic performance was assessed using suitably calibrated miniature five-hole probes as described by Wray and Carrotte [10] and employed in non-nulled mode to determine the local flow vector and total and static pressures. As indicated in Fig. 2, five-hole probe area traverses were conducted at rig inlet (X1), rotor exit (X2), OGV exit (X3), prediffuser exit (X4), within the inner (X5i), and outer (X5o) feed annuli and just upstream of the dummy injector (XI). The probes were traversed in the radial direction using a stepper motor powered linear guide attached to the external casings of the test section. Circumferential traversing was achieved by rotation of the casing or rotation of the IGV, OGV, and flametube.

Data Reduction and Errors. Overall aerodynamic performance is quantified in terms of total pressure loss and static pressure recovery coefficients derived from the five-hole probe area traverses data, corrected to the International Civil Aviation Organisation (ICAO) standard day conditions. Thus, at any measurement plane, the mass flow rate and bulk average velocity are defined:

$$\dot{m} = \int \rho U dA = \rho \bar{U} A \tag{1}$$

The mass flow balance on area traverse planes was within ±1% of the rig mass flow entering the test facility measured via a five-hole probe traverse at plane (X1). Spatially averaged values of total and static pressure at each plane were derived using the mass-weighted technique described by Klein [5], e.g.,

$$\tilde{P} = \frac{1}{\dot{m}} \int P d\dot{m} = \frac{1}{\rho \bar{U} A} \int P \rho U dA \tag{2}$$

The mass-weighted total pressure loss (λ) and static pressure rise (C_p) coefficients are then defined as

$$\lambda_{1-2} = \frac{\tilde{P}_1 - \tilde{P}_2}{\tilde{P}_1 - \tilde{p}_1} \quad C_{p_{1-2}} = \frac{\tilde{p}_2 - \tilde{p}_1}{\tilde{P}_1 - \tilde{p}_1} \tag{3}$$

The mass-weighted total pressures were repeatable to within ±1 mm H_2O, which amounts to less than ±0.5% of the dynamic pressure at OGV exit. Thus, the repeatability of the derived total pressure loss and static pressure rise coefficients was better than ±0.005 of the values presented.

Results and Discussion

Cowl Flow Variation. As an indication of the strength of the variation in cowl inlet flow conditions induced by the flow split

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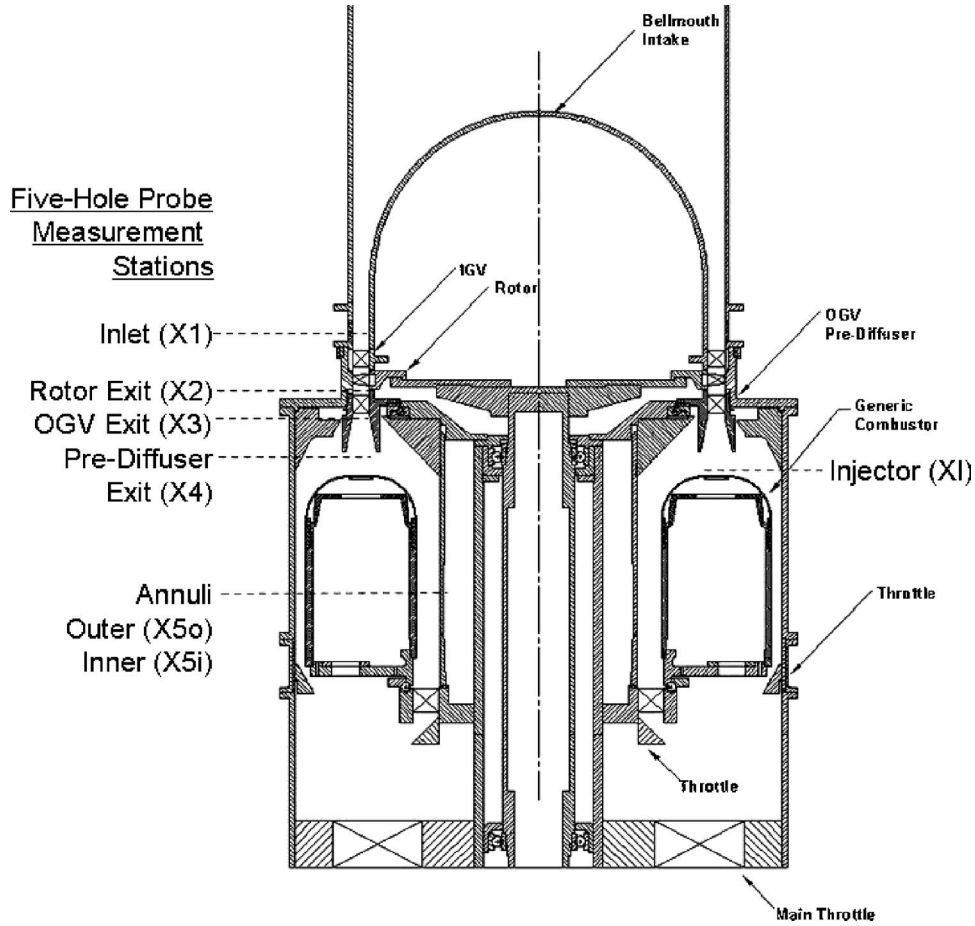

Fig. 2 Measurement planes

changes, Fig. 4 shows measured values of the total pressure field just upstream (13 mm) of the hole in the combustor cowl for the two higher mass flows (50% and 70%) (expressed as a loss coefficient with respect to the rotor exit plane (X2)). One particular point to notice is the increase in nonuniformity of the total pressure feed to the injector implied at the high flow (70%) condition. Since lean module fuel injectors may need high levels of flow uniformity to achieve lean emissions performance, it is clear that extra efforts to ensure this by suitable tailoring of the combustor external aerodynamics are likely to be needed. To increase injector flow while keeping prediffuser/dump diffuser geometry the same is unlikely to be acceptable. As the porosity of the cowl increases, and the dump gap remains the same, the downstream static pressure field as seen by the prediffuser exit changes from one that is effectively uniform circumferentially at low cowl flows—varying from a higher level in between injectors where flow impingement occurs to a slightly lower level in line with the "sink" flow areas of the cowl that feed the fuel injectors—to a larger amplitude variation in static pressure at increased cowl flow from the highs in between injector locations, to significant lows inline with injectors. At the very least, this is likely to disturb the beneficial blockage effect on the prediffuser expected at an unchanged dump gap. Whether the effect penetrates even further upstream is now addressed.

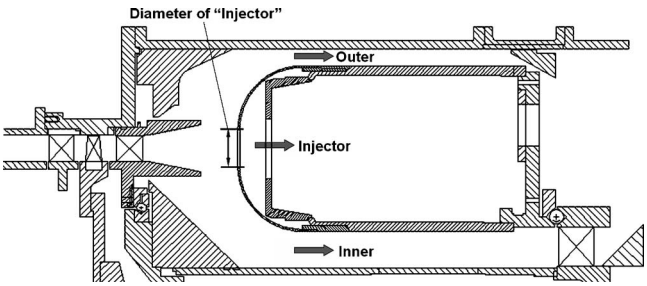

Fig. 3 Cowl details

Rig Inlet (X1) and Rotor Exit (X2). Figure 5 presents the nondimensional axial velocity profiles at rig inlet (X1) for the various cowl flow splits; the different development lengths on the inner/outer annulus walls in the rig entry section explain the different inner/outer boundary layer shapes seen at station X1. Figure 5 shows that upstream of the IGV, the inlet profile remains constant for all conditions; the IGV is too far upstream even at the highest cowl flow rate to sense the downstream change in flow split. This is, however, not the case for the rotor exit (X2) profiles, as can be seen in Fig. 6, which contains normalized and pitch-averaged axial velocity, yaw angle, and total pressure profiles; these have been averaged over a single IGV sector (i.e., two OGV sectors given the blade numbers), and the bulk Mach number at station X3 is 0.12. Cleary, for 30% and 50% cowl flow, the rotor

Table 1 Summary of test configurations

Test No.	"Injector" diameter (mm)	Mass flow split, % (inner/cowl/outer)
1	40.0	35/30/35
2	52.5	25/50/25
3	61.5	15/70/15

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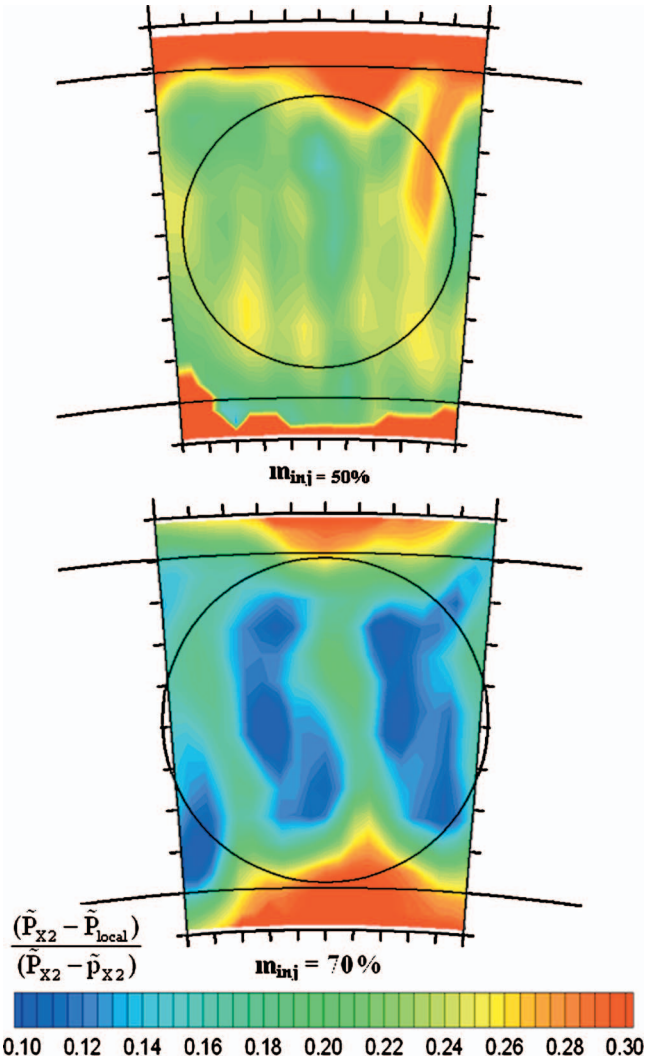

Fig. 4 Cowl inlet data: m_{inj}=50% (top), m_{inj}=70% (bottom) (height=80 mm, sector angle $\Delta\theta$=9 deg (4-OGV passages))

exit conditions only change marginally. However, for 70% of the prediffuser efflux passing through the cowl, the rotor exit flow field is clearly altered. The velocity profiles show a flow deficit at the tip of the rotor—a feature also reflected in the pressure and yaw angle profiles. The change in the combustor flow split to 70% cowl flow has created a strong enough variation in the downstream static pressure field to feed upstream and force the compressor rotor. Note that the left hand pressure profiles in Fig. 6 are plotted as two sets for the 70% case. One set represents a two OGV pitch average taken inline with injectors and one is taken in between injectors. Although some difference is seen, the nondimensionalizing dynamic head is taken relative to the local OGV sector and hence circumferential variations in flow strength are not seen. The right-hand pressure plot in Fig. 6 shows an average taken over a whole injector sector (eight OGV passages, again, one sector is in between and one inline with the injector location. Now a clear difference is seen in the total pressure level in the core passage between sectors, indicating that as the rotor passes through the circumferentially varying static pressure field, it is forced to move around on its characteristic.

Figure 7 quantifies the magnitude of the varying static pressure seen by the rotor at both 50% and 70% conditions from measurements of a fixed wall static at rotor exit (X2) as the combustor was rotated though two sectors. At 50% cowl flow, there are only slight variations in the measured pressure circumferentially, probably due to the potential field of the OGV, but only just within measurement accuracy. The static pressure for the 70% case, however, can be seen to vary significantly from inline to in between injectors, showing a low pressure inline. The circumferential resolution in the data is again just sufficient to pick up the changes in static pressure caused by the potential field of the downstream OGVs. However, this is small compared to the variation caused by the cowl flow suction field, which is equivalent to approximately 5% of the rotor total pressure rise. This is certainly enough to force the rotor to move locally up or down its characteristic. The fact that the changes in downstream flow split can penetrate as far upstream as the rotor and can affect its operation is significant, and shows that dump gap design rules to achieve optimum stability for prediffusers will clearly not be transferable between conventional and lean module combustors.

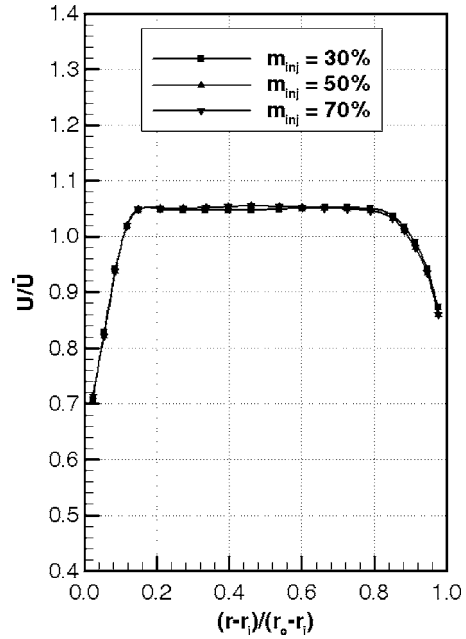

Fig. 5 Rig inlet axial velocity profile (X1)

OGV Exit (X3). Final confirmation of the change in upstream interaction between 50% and 70% flow split is provided in Fig. 8. This shows the circumferential variation of a single OGV exit (X3) total pressure measurement located at midannulus height between two OGVs as the combustor is rotated through one injector sector. At 30–50% cowl flow, the pressure is constant to within measurement accuracy. Caution must be exercised here in that this measurement is only at midblade passage, midannulus height, so it cannot be unequivocally concluded that the OGV row does not see the variation in circumferential blockage provided by the combustor (see below). However, at 70%, the measured pressure at OGV exit varies significantly and periodically with the injector location. The magnitude of this variation is relatively large at 7% of the local dynamic pressure. Velocity data averaged over respective sectors shows that with respect to the rig overall mass flow rate, the indicated mass flow per sector varies from inline to in between injectors by ±5%. It is clear, therefore, that at 70%, the downstream pressure field has propagated upstream and beyond the OGV to influence the rotor. The OGV performance is also affected by this interaction. Figure 9 plots axial velocity contours for two 2-OGV sectors in between and inline with injectors. Adjacent OGV wakes differ because there is one IGV per two OGVs and the IGV wake persists through the rotor, deepening the left-hand OGV wake. For 70% cowl flow, there appears to be a reduction in the bulk velocity. Table 2 details the total pressure loss and static pressure recovery across the OGV. For the two lower flow

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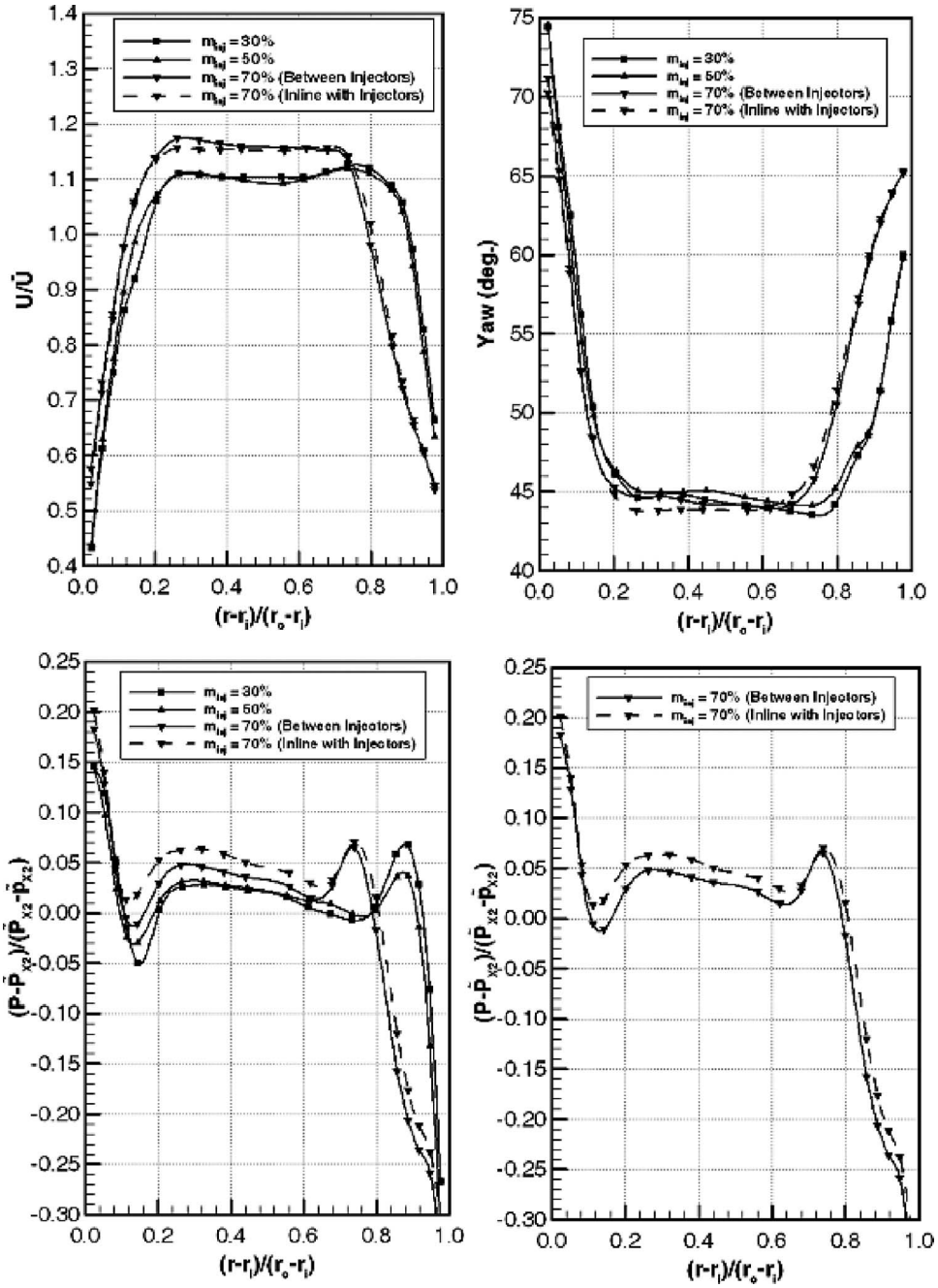

Fig. 6 Pitch averaged profiles at rotor exit (X2)

splits, performance is maintained. Over a whole injector sector it would appear that 70% cowl flow produces a similar level of OGV performance. However, if separate 2-OGV sectors are considered, this is not the case. Between injectors, where there is a flow deficit, λ increases from 0.13 to 0.15, whereas inline with injectors, where there is increased flow, λ falls to 0.1.

Prediffuser Exit (X4). Given that, for Test 3, the rotor and OGV are affected by the circumferential variation in blockage presented by the combustor, it would be logical to expect this to be reproduced at prediffuser exit. Additionally, it would also be reasonable to assume that variations of prediffuser exit flow should be observed as the airflow through the cowl increases. Figure 10 indicates that this is indeed the case. Normalized axial velocity contours were gathered over a full injector sector (the bulk average Mach number at station X4 is 0.07). Clearly, as the combustor mass flow is redistributed, the flowfield and OGV wake structure also change. Increasing the injector flow from 30% to 50% alters the OGV wake development, with the flow taking on an inboard bias. The most likely explanation for this is that at 50% flow, although the upstream interaction has not reached the rotor, it has reached the OGV and the total pressure profile at OGV exit has been modified. This change is then amplified by the prediffuser to produce the inboard bias shown. At 70% cowl flow, the flow structure at prediffuser exit changes again, exhibiting low velocity regions suggesting the inner wall boundary layer in particular is under duress. The residual swirl level at prediffuser exit also changes with test condition. At 30% cowl flow, the pitch-averaged residual yaw angle is ~−4 deg over the central 60% of the passage height, rising to ~3 deg near the diffuser end walls. At 50% flow, this changes to ~4 deg in the core flow and as much

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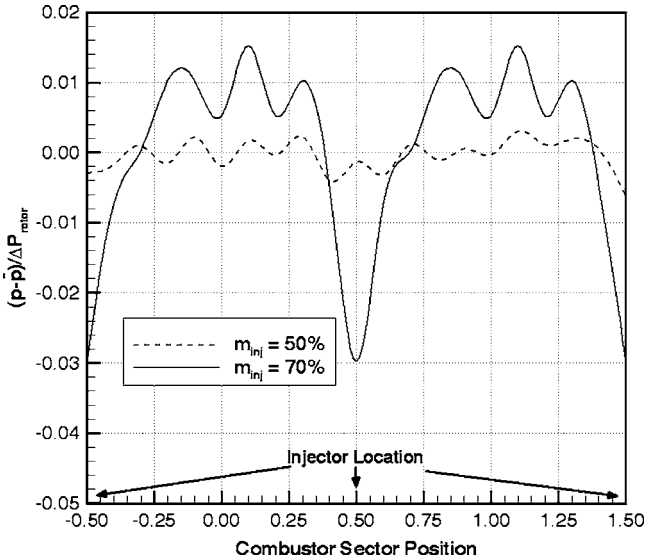

Fig. 7 Rotor exit (X2) static pressure

as 20 deg in the wall regions. At 70% cowl flow, since the rotor exit condition has now changed appreciably, the core swirl angle is ~−2 deg over most of the passage height, rising to ~5 deg near the outer wall.

Overall System Performance. Measurements in both feed annuli allow the overall system performance to be established. Table 3 gives the mass-weighted total pressure loss and static pressure recovery coefficients for the inner annulus (X5i) quoted with respect to both rotor exit (X2) and OGV exit (X3). As the injector mass flow is increased from conventional air-spray levels (30%), to levels more typical of lean module designs (50–70%), the total pressure loss increases. The system loss from rotor exit is 22% of rotor exit dynamic for 30% cowl flow and this rises to 23% for 50% and 26% for 70%. The overall system static pressure rise is not significantly affected, but both λ and C_p values from OGV exit to annulus show a large increase in loss and deterioration in feed quality as the injector mass flow increases. The data for the outer annulus indicate similar trends and are not repeated here.

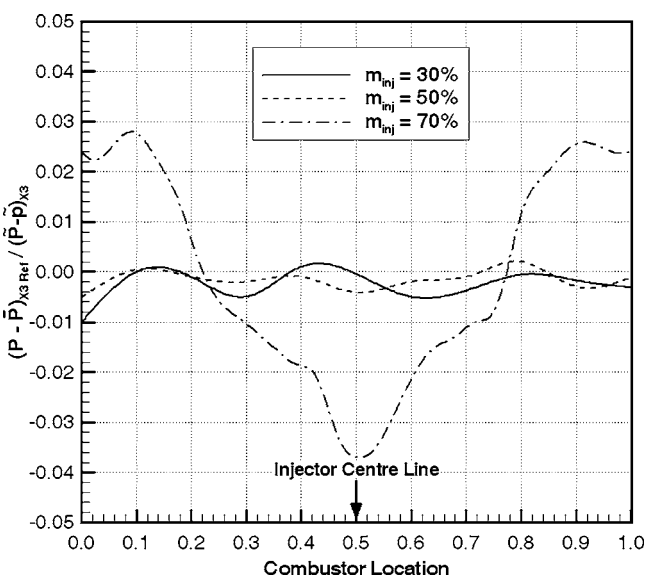

Fig. 8 OGV exit (reference) total pressure

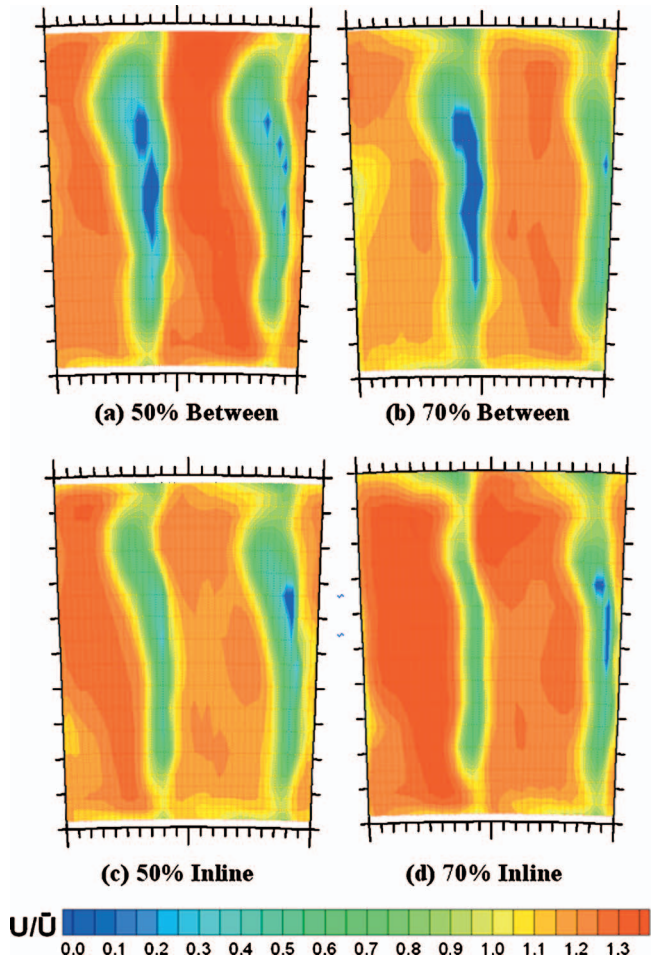

Fig. 9 Axial velocity contours at OGV exit (X3): (*a*) 50% between, (*b*) 70% between, (*c*) 50% inline, (*d*) 70% inline (height =36.6 mm, sector angle $\Delta\theta$=4.5 deg (2-OGV passages))

Conclusions

The effect on combustion system external aerodynamics of increasing the injector flow split from air-spray levels to those typical of lean module injectors has been studied. In order to do this, three combustor flow configurations were tested each utilizing the same inlet section, IGV, rotor, and integrated OGV/prediffuser. The flow fraction of compressor efflux entering the cowl was systematically increased from 30% to 50% and 70% while increasing the effective area of cowl to maintain a constant mass flow per unit area. From the results presented here, the following conclusions can be made:

- For all flow splits, rig inlet conditions remained constant.
- At rotor exit, there was no change in the flowfield as the cowl mass flow increased from 30% to 50%. At 70%, a notable circumferential variation in the static pressure field

Table 2 OGV performance data

Test No.	λ_{X2-X3}	Cp_{X2-X3}
1	0.132	0.38
2	0.128	0.38
3 (Burner sector)	0.129	0.38
3 (2-OGV sector between)	0.153	0.36
3 (2-OGV sector inline)	0.097	0.39

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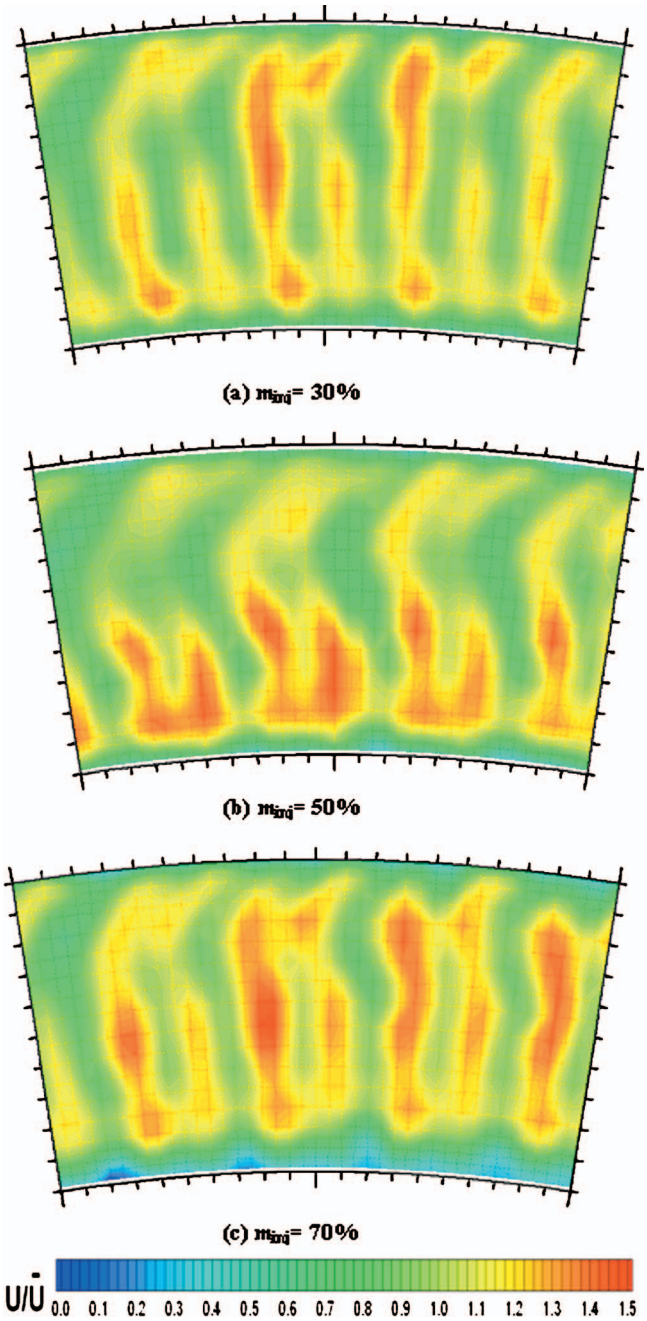

Fig. 10 Axial velocity contours at prediffuser exit (X4): (*a*) m_{inj}=30%, (*b*) m_{inj}=50%, (*c*) m_{inj}=70% (height=65.9 mm, sector angle $\Delta\theta$=18 deg (8-OGV passages))

was observed. This resulted in a degradation of the outboard, tip flow as the rotor was forced locally to move up/down its characteristic.

- Similarly for 30% to 50% cowl flow, the performance of the OGV row did not change, but at 70%, a circumferential redistribution of mass flow was observed, increasing by ~5% in line with the injectors and decreasing by the same amount in between. Consequently, the OGV total pressure loss coefficient varied from 0.097 in line to 0.153 between injectors with an overall sector average of 0.129.
- As the cowl mass flow increased from 30% to 50% and 70%, the prediffuser flow quality worsened, particularly near the inner wall, where at 70% the prediffuser was close to separation.
- System performance from rotor through to feed annuli is a direct function of the condition of the prediffuser exit flow field, particularly the boundary layer flow. As the injector air flow was increased and the prediffuser performance decreased, the annulus flow incurred higher losses and became less uniform. For both annuli the loss coefficient increased, suggesting an increase in system loss of almost 20% at the higher injector flow.
- An estimate of injector feed suggested that at the higher flow rate, the total pressure was nonuniform.

Table 3 Inner annulus performance data

Test No.	λ_{X2-X5I}	λ_{X3-X5I}	$C_{p_{X2-X5I}}$	$C_{p_{X3-X5I}}$
1	0.224	0.159	0.71	0.71
2	0.232	0.204	0.72	0.69
3	0.262	0.284	0.71	0.65

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Nomenclature

A = passage area
AR = area ratio
c = outlet guide vane chord
C_p = static pressure rise coefficient
h = annulus passage height
L = diffuser length
$\dot{m}$ = mass flow rate
N = rotor speed
P,p = local total or static pressure
R = gas constant (perfect gas law)
R_n = nondimensional radius $((r-r_i)/(r_o-r_i))$
r = radius relative to rig centerline
r_i, r_o = inner casing, outer casing radius
s = blade pitch
T,t = total or static temperature
U = rotor midpassage blade speed
V_a = midpassage axial velocity
x = axial distance
α = kinetic energy flux coefficient
ϕ = flow coefficient (V_a/U)
ρ = density
λ = total pressure loss coefficient

Subscripts

X1 = rotor inlet traverse plane
X2 = rotor exit/OGV inlet traverse plane
$X3$ = OGV exit traverse plane
$X4$ = prediffuser exit traverse plane
$X5i$ = inner annulus traverse plane
$X5o$ = outer annulus traverse plane

Superscripts

$-$ = area weighted spatially averaged mean value
$\sim$ = mass weighted spatially averaged mean value

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